

Topic: Using Amazon RDS (SQL Server Express) with QuickSight

✓ Why do we use this? (Purpose)

- To **store data online** → using Amazon RDS
- To **manage data easily** → using SQL Server
- To **analyze and show data in charts** → using Amazon QuickSight

👉 In short:

Store → Manage → Visualize data

⚙️ How it works (simple explanation)

1 Create database (Amazon RDS)

- Go to AWS → RDS
 - Create database using SQL Server Express
 - Set username, password
 - You get an **endpoint (server address)**
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2 Create table & insert data

Use:

👉 SQL Server Management Studio

- Connect using endpoint
 - Create table
 - Insert data using SQL queries
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3 Connect to QuickSight

- Open QuickSight
- Click **New Dataset** → SQL Server
- Enter:
 - Endpoint
 - DB name
 - Username & password

Visualize data

- Select table
- Create charts (bar, pie, etc.)
- Build dashboard

In one line (very important for viva)

We use Amazon RDS to store data, SQL Server to manage it, and QuickSight to visualize it in charts and dashboards.

Extra points (to impress examiner)

- It is **cloud-based**  (no need to install database locally)
- Easy to **scale and access from anywhere**
- Used in **business analytics and reporting**

Very short answer (if examiner asks quickly)

Amazon RDS stores the database, SQL Server is used to create and manage tables, and QuickSight connects to the database to create visualizations like charts and dashboards.

Topic: Setup, create & connect a WordPress site to object storage using Lightsail

Why do we use this? (Purpose)

- To **host a website easily** → using WordPress
- To **store images/files separately** → using object storage (bucket)
- To **improve speed and performance** 
- To **save server space and cost**

 In short:

Website (Lightsail) + Storage (Bucket) = Fast & scalable website

How it works (simple explanation)

1 Create WordPress site (Lightsail)

- Go to AWS Lightsail
- Click **Create Instance**
- Choose:
 - Platform → Linux
 - Blueprint → WordPress
- Launch instance

👉 Your website is ready with a public IP

2 Create Object Storage (Bucket)

- In Lightsail → go to **Storage** → **Buckets**
- Create a new bucket (give name)

👉 This bucket will store images, videos, files

3 Connect WordPress to Bucket

- Open your WordPress admin panel
- Install a plugin like:
 - WP Offload Media
- Configure plugin:
 - Enter bucket name
 - Add access keys

👉 Now media files go directly to bucket

4 Use and verify

- Upload image in WordPress
- It gets stored in bucket instead of server

In one line (very important for viva)

We use Lightsail to host WordPress and connect it to an object storage bucket to store media files separately for better performance and scalability.

Extra points (to impress examiner)

- Reduces **server load** ⚡
 - Improves **website speed**
 - Easy to **scale storage**
 - Cost-effective
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Very short answer

WordPress is hosted on Lightsail, and an object storage bucket is created to store media files. A plugin connects WordPress to the bucket so files are stored externally.